

Jul 03, 2026

Introduction to Computer and Network Security

Part 1

You have been employed by a news agency that is in the process of introducing an AI-powered chatbot that can address readers' inquiries. The objective of this chatbot is to assist readers in comprehending the technical aspects and the overall impact on their security and privacy posture. You are expected to help them assess the quality of the answers.

Question 1 [35 points]

You ask the LLM to describe how **Asymmetric Encryption** works. It answers with:

Asymmetric Key Encryption is a hashing method where three entities use a common private key to communicate securely. The sender uses the key to encrypt the ciphertext, and the receiver uses the same key to decrypt the message. Both the sender and the receiver use the same key to encrypt and decrypt messages, but only one of them keeps the private key secret. This maintains confidentiality.

This mechanism is significantly faster than Symmetric Encryption.

Describe the errors and provide a correct and more comprehensive version of the response.

Question 2 [35 points]

You ask the LLM to describe what **Multi-factor authentication** is and what are the possible classes of available authentication factors. It answers with:

Multi-factor Authentication (MFA) is a security mechanism that requires users to provide one or more independent factors to verify their identity before being granted access to a system, application, or data. The goal of MFA is to simplify access control and reduce the risk of authorized access, especially in cases where a single authentication method (like a password) might be compromised.

MFA uses a combination of different authentication factors, which fall into five main categories:

- Something you know – e.g., a password or PIN.

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- Something you have – e.g., a smartphone, smart card, or security key.
- Something you do – e.g., solve a CAPTCHA, record a video showing your ID and reading a text on the screen.
- Something you ask – e.g., a one-time code sent via SMS.
- Something you are – e.g., a fingerprint, facial recognition, or iris scan (biometric data).

Describe the errors and provide a correct and more comprehensive version of the response.

Part 2

Question 3 [10 points]

Define the notions of risk and vulnerability and provide (at least) an example for each one.

Question 4 [10 points]

Briefly explain the Zero Trust security model and describe a scenario in which it can be beneficial.

Question 5 [10 points]

Briefly describe OAuth 2.0 and discuss whether authenticating with it is a good or bad idea.

Please, remember...

- on each used sheet of paper (possibly, 2 sheets at most), annotate:
 - first and last name
 - student ID number
 - question number
- write either in Italian or English, acronyms do not need to be translated;
- only pens are allowed. Drafts and text written using pencils will not be considered during the exam revision;
- adequately justify your answer. A single phrase and a paragraph-long answer will have a significantly different grade;
- write clearly, your answers will not be *interpreted*;
- after turning in your exam, please respect your classmates by leaving quietly.